

IE7615 Project 1 Proposal

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Celebrity Selection and Identities

For the initial phase of the CelebA classification pipeline, I have selected a subset of five distinct identities. My selection process was governed by three primary criteria: visual diversity, demographic representation and dataset balance.

- **ID: 7** - A. R. Rahman (24 images)
- **ID: 797** - Aria Crescendo (25 images)
- **ID: 2619** - Dmitry Medvedev (25 images)
- **ID: 4428** - Jenna Fischer (23 images)
- **ID: 7007** - Monika Brodka (24 images)

The chosen identities represent a wide spectrum of facial features to ensure the model learns robust, generalised characteristics rather than specific demographic biases. By including individuals with varying hair colors (ranging from blonde to black), different skin tones and diverse facial structures (encompassing a range of masculine and feminine features), we provide the network with a high-contrast training environment. This diversity can be crucial for the model to learn "face-specific" features, such as eye shape and bone structure, rather than relying on easy-to-memorize features like hair color or skin tone.

Framework Choice

CNN

This architecture employs a four-block convolutional hierarchy, progressively increasing filter depth from 16 to 128. It integrates aggressive spatial augmentations with heavy regularization, specifically L2, Dropout, and Batch Normalization to prevent overfitting. By utilizing Global Average Pooling instead of traditional flattening, the model maintains a lightweight, parameter-efficient design optimized for robust feature extraction on highly constrained datasets.

ResNet

Utilising transfer learning, this model repurposes a pretrained ResNet50 backbone for feature extraction. The pipeline incorporates intensive augmentation and normalisation layer for ResNet specific preprocessing. By appending a regularized 256-unit dense head with 50% Dropout, the architecture aims to transfers robust ImageNet learned weights while preventing over-fitting on our highly constrained celebrity dataset.

ResNet with Fine Tuning

Building on the feature-extraction baseline, this model employs partial unfreezing to adapt ResNet50's upper layers to my limited celebrity data domain. By combining intensive data augmentation with a regularized 256-unit dense head and Global Average Pooling, the architecture seeks to capture domain-specific features while preventing the catastrophic forgetting or overfitting typically associated with fine-tuning on small datasets.

Division of Labour

As this project has been undertaken by myself, rather than as a group, all of the work was done by myself.

Anticipated Risks

The primary constraint of this study is the extremely small dataset size (121 images total), which introduces a significant risk of both overfitting and underfitting. The high intra-class variance within the subset—specifically the wide range of hairstyles, hair colours, and makeup in the Monika Brodka sample, makes it difficult for models to learn stable, consistent patterns.

To mitigate this, a stratified 70/15/15 split was implemented per celebrity to ensure each split contains a representative sample. However, with a test set of approximately 20 images, the results will be subject to high statistical volatility; a single misclassification induces a ~5% swing in accuracy, which may manifest as erratic validation curves. Consequently, while the custom CNN may produce volatile results due to its limited training data, the ResNet models which leverage pre-existing knowledge are expected to provide more stable and superior performance.